



- 6 (a) (i) State what is meant by rectification of an alternating voltage.

.....
 [1]

- (ii) State the difference between half-wave rectification and full-wave rectification.

.....

 [2]

- (b) (i) Complete Fig. 6.1 to show a circuit that produces half-wave rectification of an alternating input voltage V_{IN} to produce output voltage V_{OUT} across the resistor R .

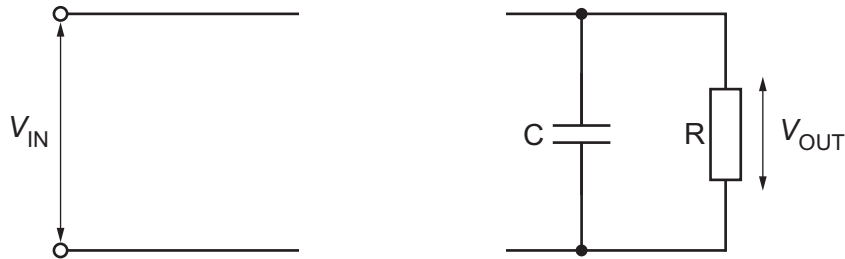


Fig. 6.1

[2]

- (ii) State the purpose of the capacitor C in the circuit of Fig. 6.1.

.....
 [1]

- (c) The input voltage V_{IN} in Fig. 6.1 is a square wave. Fig. 6.2 shows the variation of V_{IN} with time t .

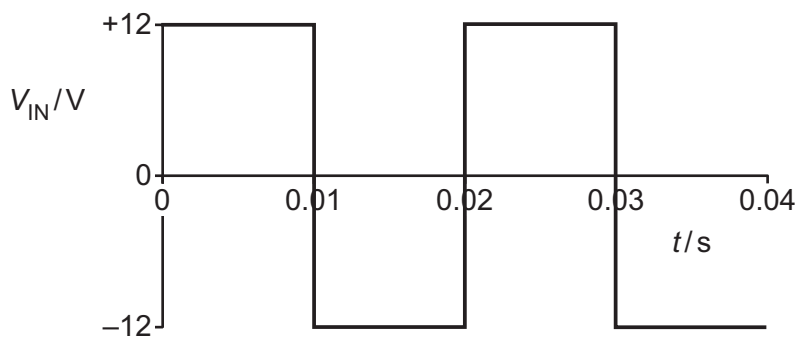


Fig. 6.2



Fig. 6.3 shows the variation of V_{OUT} with t .

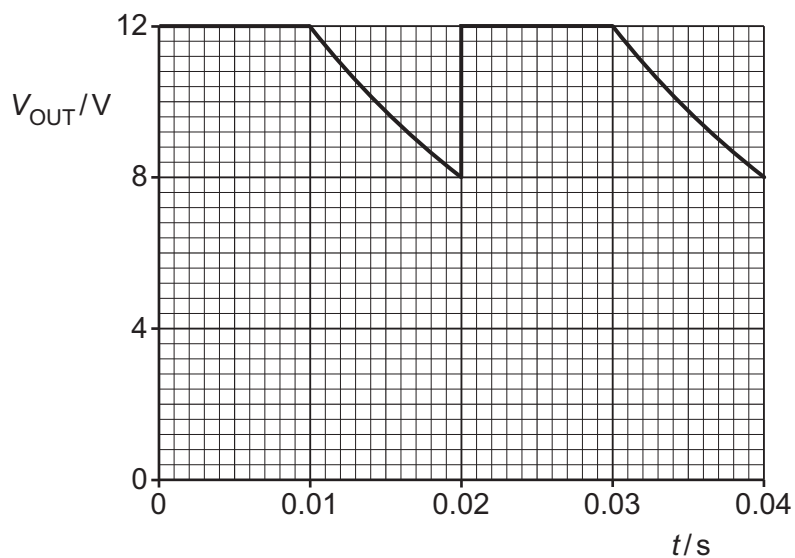


Fig. 6.3

The maximum energy stored in the capacitor is 0.041 J.

(i) Show that the capacitance of C is $570 \mu\text{F}$.

[2]

(ii) Determine the resistance of R .

resistance = Ω [3]

[Total: 11]

